

Radar-based Sentiment Analysis and Recognition

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Abstract

1 Introduction

Sentiment analysis commonly refers to the process of identifying and extracting subjective information from text, speech, or other forms of data [Ain *et al.*, 2017]. In this study, we focused on sentiment analysis using radar-based data. Radar sensors are a powerful tool for capturing subtle movements and gestures. They are non-intrusive, privacy-preserving, and much easier to deploy in real-world scenarios compared to other modalities such as cameras or wearable sensors [Sun *et al.*, 2020]. The data from these sensors have proven useful in applications such as gesture recognition [Sun *et al.*, 2020], object detection [Manjunath *et al.*, 2018] and human activity recognition [Froehlich *et al.*, 2024].

The main focus of researchers in the field of sentiment analysis has been on text-based and speech-based data, with a significant amount of work done on NLP models [Rokade *et al.*, 2025] and speech processing techniques. The objective was always to create models that understand human emotions through media but rarely through their body movements. Consequently, the use of radar-based data for sentiment analysis and recognition is still a relatively unexplored area, with limited research and few datasets available. The characteristics of radar signals, such as their ability to capture micro-movements and gestures, make them a promising source of information for micro-movement based experiments such as sentiment analysis [Sun *et al.*, 2020].

The objective of this paper is to explore the potential of radar-based data for sentiment analysis and recognition. We propose different approaches to transform radar signals into training data and evaluate how effective the classification models are in recognizing different sentiments. We split the paper into three experiments, each focusing on a different aspect of the problem. The first experiment focuses on feature-based classification using a mix of kinematic, behavioral and spatial features extracted from the radar signals. The second experiment is based on transformers [FILL HERE]. The third is [FILL HERE].

2 Related Work

This research is based on the experiments conducted by [Zuo *et al.*, 2025], in which they introduced "a multimodal radio frequency dataset" for both human behavior recognition and emotion analysis. By conducting a series of experiments on 44 participants, they collected data from 13 radars, 8 RFID tags, LoRa devices and infrared cameras in 4 different campaigns that each focused on a different aspect of human behavior. We will be interested in this paper on the third campaign that was built around 6 sentiment states: *depression, distraction, excitement, focus, relaxation and stress*.

3 Experiment 1: Feature-Based Classification

3.1 Preprocessing

The data is currently composed of a sequence of XYZ coordinates representing the position of a target in 3D space over time. The first step in preprocessing is to synchronize the data from different radar sensors to ensure that the data from different sensors is aligned in time, as this will allow for accurate feature extraction. Once the data is synchronized, the data is normalized on a user basis so the user-specific biases are removed.

With the data synchronized and normalized, we can proceed to segment the data into smaller windows. This is done to capture the temporal dynamics of the target's movement. The window size and stride are hyperparameters that can be tuned based on the specific application and dataset. In this study, we experimented with different window sizes and strides to find the optimal configuration for our classification task. Lower window sizes tend to underperform and higher window sizes tend to overfit the data and lose generalization. The optimal window size was found to be 90 frames with a stride of 20 frames. This configuration allows for capturing the temporal dynamics of the target's movement while also providing enough data for training the classification models.

3.2 Feature Extraction

Given the preprocessed data, we can extract a set of features that capture the kinematic, behavioral and spatial characteristics of the target's movement. The features are extracted from each window of data and are used as input to the classification models. The features are divided into three categories: motion dynamics, statistical and directional features.

Feature	Description
Path Efficiency	Ratio of displacement to trajectory length
Mean Speed	Average speed.
Std. Speed	Variability in speed.
Median Speed	Median speed.
25th Percentile Speed	Lower quartile of speed.
75th Percentile Speed	Upper quartile of speed.
Mean Acceleration	Average rate of change in speed.
Std. Acceleration	Variability of acceleration.
Median Acceleration	Median acceleration.
25th Percentile Acceleration	Lower quartile of acceleration values.
75th Percentile Acceleration	Upper quartile of acceleration values.

Table 1: Motion dynamics features.

Feature	Description
Mean Jerk	Average change of acceleration
Std. Jerk	Variability in jerk values.
Median Jerk	Median jerk.
25th Percentile Jerk	Lower quartile of jerk values.
75th Percentile Jerk	Upper quartile of jerk values.
Spectral Centroid	Center of mass of the movement frequency spectrum.
Spectral Entropy	Complexity of the frequency spectrum.
Speed CV	Normalized measure of speed variability.
Direction Change Rate	Rate of changes in movement direction.
Speed Autocorrelation	Temporal dependence of cursor speed.
Tortuosity	Degree of path curvature relative to its overall length.

Table 2: Statistical and directional features.

Feature	Description
Speed Skewness	Asymmetry of the speed distribution.
Fraction Stationary Time	Proportion the target remains stationary.
Mean Turn Angle	Average angle between consecutive movement segments.
Turn Angle Entropy	Diversity and unpredictability of turning angles.
Explained Variance Ratio 1	Variance explained by the first principal component of the trajectory.
Explained Variance Ratio 2	Variance explained by the second principal component.
Explained Variance Ratio 3	Variance explained by the third principal component.

Table 3: Statistical and directional features.

Algorithm 1 Feature Selection and LOSO Classification

```

for each window configuration  $(w, s)$  do
  for each feature count  $k \in K$  do
    for each Leave-One-Subject-Out split do
      Split the data into training and testing sets.
      for each radar  $r = 1, \dots, 13$  do
        Select the top- $k$  features using the ANOVA F-test.
        Transform the training and testing feature sets.
      end for
      Concatenate the selected features from all radars.
      for each classifier  $m \in \mathcal{M}$  do
        Train classifier  $m$ .
        Evaluate on the test set.
        Record the accuracy and weighted F1-score.
      end for
    end for
    Compute the mean accuracy and weighted F1-score across all folds.
  end for
end for
for each classifier do
  Select the configuration with the highest mean accuracy.
end for
Output: Best window size, stride, feature count, accuracy, and weighted F1-score for each classifier.

```

80 The motion dynamics features capture the speed, acceleration
81 and jerk of the target’s movement. The statistical features
82 capture the distribution of the target’s movement in terms of
83 speed, acceleration and jerk. The directional features capture
84 the directionality of the target’s movement in terms of turn
85 angles and explained variance ratios. The extracted features
86 are summarized in Tables 1, 2 and 3.

3.3 Training Loop

88 With the current setup, the data is now composed of 27 fea-
89 tures per radar sensor. Given that we have 13 radar sensors,
90 the total number of features is 351 over more than 600 win-
91 dows. For an effective training of the classification models,
92 we need to reduce the dimensionality of the data and select
93 the most relevant features. To find the optimal configuration
94 of window size, stride and number of features, we imple-
95 mented a training loop that iterates over different configura-
96 tions and evaluates the performance of the classification mod-
97 els using Leave-One-Subject-Out (LOSO) cross-validation.
98 It uses the ANOVA F-test to select the top- k features for
99 each radar sensor and concatenates them to form the final

feature set. The classification models are then trained on the
training set and evaluated on the test set. The accuracy and
weighted F1-score are recorded for each configuration and
the best configuration is selected based on the highest mean
accuracy across all folds. The training loop is summarized in
Algorithm 1.

3.4 Results

The best configuration for each model is selected based on the
highest mean f1-score across all folds. As previously men-
tioned, the most optimal configuration across all models was
found to be a window size of 90 frames, a stride of 20. The

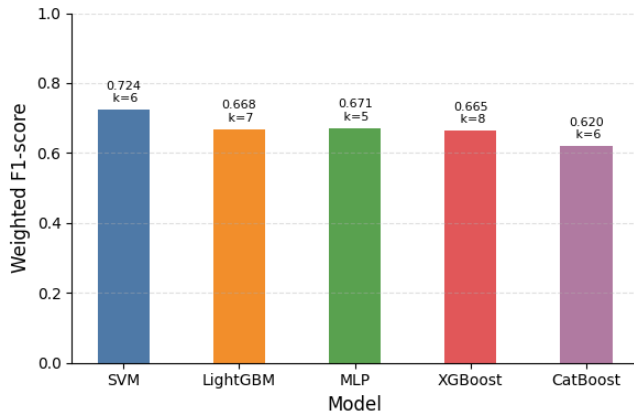


Figure 1: Models comparison based on F1 score.

results of the training loop in Figure 1 are based on models trained with this configuration.

It shows that the SVM model outperforms the other models with a mean f1-score of 0.72, followed by LightGBM, xGBoost and MLP achieving mean f1-scores in the range of 0.66 and 0.67. The CatBoost classifier underperforms with a mean f1-score of 0.62.

4 Experiment 2

5 Experiment 3

6 Results

7 Discussion

Acknowledgments

References

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